Modelling and Switching Control of Dynamical Systems

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Nomenclature

$\eta \in (0, 1)$	=	The efficiency of the solar panel
ω_0	=	Flow rate in the hydraulic system
ω_1	=	Flow rate in the heating/cooling system
θ	=	Angle of window opening
$A_c,\ A_T,\ A$	=	Area of the piston of the hydraulic cylinder, area of Tank 2, area of the solar panel.
c	=	Specific heat capacity of the fluid
$C_r,\ C_2,\ C_B$	=	Heat capacity of the room, radiator and boiler
F_c	=	Force of the hydraulic cylinder
h	=	Water level height in Tank 2
$I(t)$	=	The irradiance of the solar panel (W/m2)
m_{flu}	=	Mass of the fluid
R_0, R_r, R_B	=	Thermal resistance of the room, outside environment and boiler.
R_1	=	Resistance of resistor 1 in the electrical circuit
$T_0,\ T_1,\ T_2,\ T_B,\ T_r$	=	Temperature of the environment, cooling liquid, radiator, boiler and room

Introduction

This is the report for the mini project of Modelling of Dynamical Systems. The report is divided into each sections for each seperate question, with subsections corresponding to the subquestions.

I. Physical Modelling

1. Heating mode

With the heating mode turned on, and the window completely closed, we can derive a set of differential equations on how the solar panel, boiler and radiator effect the room temperature. Using the formulas in figure 2.5 from the reader, we can obtain a model. First, a model is made for how the solar panel is effecting the boiler temperature:

$$\dot{T}_B C_B = I(t) A \eta \frac{R_B}{R_B + R_1} \tag{1}$$

For the $I(t)$ the first half of a sign function that starts at 6:00 and ends at 20:00 with an amplitude of half of the maximum solar radiation on Earth. Then, the interaction between the radiator and the room can modelled:

$$\dot{T}_2 C_2 = \frac{1}{R_2}(T_r - T_2) - T_2 f_1 \tag{2}$$

$$\dot{T}_r C_r = \frac{1}{R_2}(T_2 - T_r) + \frac{1}{R(\theta)}(T_0 - T_r) \tag{3}$$

$$\tag{4}$$

Since heating mode is switched on, warm water from the boiler will flow in the radiator. Since equation 3 already incorporates the temperature of the radiator, it does not need additional terms. The temperature of the boiler is effected, since cooled down water is flowing back from the radiator. The radiator temperature is of course effected as well, since hot water is flowing in from the boiler.
These terms that depend on the mode of the system will be denoted by Δ in the report.

$$\Delta\dot{T}_2 C_2 = T_B f_1 \tag{5}$$

$$\Delta\dot{T}_B C_B = (T_2 - T_B) f_1 \tag{6}$$

$$f_1 = \dot{m} c \tag{7}$$

Of course, $\dot{m} = \omega_1$. With these equations, we can describe the heating of the room.

2. Bond graph

For the completed bond graph, check appendix A.

3. 1.3 - Model of complete system

To describe the model of the complete system, all the modes are separately defined. On these modes, we have three subsystems that can be combined: the boiler, the radiator, and the room with the hydraulic window. These systems have effects on each other and the states are depending on the positions of the valves controlling the heating and cooling system.

1. Heating system

For the heating system the equations are already derived in section I.A. The results are shown in Equation 1. The cooled fluid that comes back from the radiator influences the boiler temperature, which is captured in Equation 6.

2. Cooling system

For the cooling system, the radiator is filled with cold fluid from tank 1. This water is pumped trough the radiator, after which it is discharged in tank 2. The fluid outflow is restricted by a hole with a flow resistance of R_T and, therefore the fluid level will rise. Since the heating and cooling system can never be turned on at the same time, the equation for the temperature in the radiator equals:

$$\Delta \dot{T}_2 C_2 = T_1 f_1 \tag{8}$$

The water level in the tank is influenced by two things: outflow of the cooling system, ω_1, and the pressure that is created by the water level in the tank.

$$\dot{h} = -\frac{1}{A_T}\left(\frac{1}{R_T} h\rho g\right) \tag{9}$$

$$\Delta \dot{h} = \frac{1}{A_T}\omega_1 \tag{10}$$

3. Window

The last part of the system that needs to be modelled is the window. The window can be opened by pumping fluid into the hydraulic cylinder, by turning valve 0 into the opening mode. The fluid is pumped into the cylinder with rate ω_0. We consider the pump powerful enough that the weight of the window does not have an influence on the opening speed, and that the flow of the pump directly accelerates the cylinder.

Using equations from figure 2.14 in the reader, we can model the velocity of the endpoint of the hydraulic cylinder.

$$\omega_0 = A_c \dot{x}_c \tag{11}$$

We use the geometric properties of the window-cylinder construction. Since it is an equilateral triangle, we can model the opening of the window as:

$$\sin(\frac{\pi}{4} + \frac{\theta}{2}) = 2\frac{x_c}{l} \tag{12}$$

$$\dot{\theta} = \frac{4\omega_0}{A_c \cos\left(\frac{\pi}{4} + \frac{\theta}{2}\right) l} \tag{13}$$

The modelling of the closing of the window is a bit more complicated. Valve 0 switches positions to connect the hydraulic cylinder to the outflow tube with resistance R_0. The weight of the window forces the liquid out of the cylinder and closes the window.

The closing of the window can be modelled as a pendulum, which experiences a damping force of the hydraulic cylinder.

$$\Sigma M = \ddot{\theta} I \tag{14}$$

$$\Sigma M = -\frac{l}{2} mg \sin\theta + F_c \sin(\frac{\pi}{4} - \frac{\theta}{2}) \tag{15}$$

The force in the cylinder is depending on the resistance in the outflow tube, R_0. We use equation 11 as well as the equation from figure 2.4 from the reader:

$$\omega = \frac{1}{R1}(P_1 - P_2) = \dot{x}_c A_c \tag{16}$$

$$\left(\frac{F_c}{A_c} - P_0\right)\frac{1}{R_0} = \dot{x}_c A_c \tag{17}$$

$$F_c = (R_0 \dot{x}_c A_c + P_0) A_c \tag{18}$$

So with all these equations in place, we can combine equation 18 and 15. This gives:

$$\begin{aligned} \ddot{\theta} I &= -\frac{l}{2} mg \sin\theta \\ &\quad + (R_0 \dot{x}_c A_c + P_0) A_c \sin(\frac{\pi}{4} - \frac{\theta}{2}) \\ \dot{x}_c &= \frac{1}{4} l \cos(\frac{\pi}{4} + \frac{\theta}{2})\dot{\theta} \end{aligned} \tag{19}$$

Combining everything we get:

$$\begin{aligned}\ddot{\theta} I = &-\frac{l}{2} mg \sin\theta \\ &+ (A_c^2 R_0 \frac{1}{4} l \cos(\frac{\pi}{4} + \frac{\theta}{2})\dot{\theta} \\ &+ A_c P_0) \sin(\frac{\pi}{4} - \frac{\theta}{2})\end{aligned} \tag{20}$$

4. State space system

To capture this model in a state space system, a system of nonlinear differential equations is made. Because there are different modes in the system, discrete variables are used to turn the modes on and of. The terms that are changing in that case are denoted by Δ.

$$\begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} \left(\frac{1}{R_2}(T_2 - T_r) + \frac{1}{R(\theta)}(T_0 - T_r)\right)/C_r \\ \left(\frac{1}{R_2}(T_r - T_2) - T_2 f_1\right)/C_2 \\ \left(I(t) A \eta \frac{R_B}{R_B + R_1}\right)/C_B \\ 0 \\ 0 \\ -\frac{1}{A_T}\left(\frac{1}{R_T} h \rho g\right) \end{pmatrix} + \Delta \tag{21}$$

1. Heating

When the heating mode is turned on, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ (T_2 - T_B) f_1 / C_B \\ (T_B f_1)/C_2 \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{22}$$

2. Cooling

When the cooling mode is turned on, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ T_2 f_1 / C_2 \\ 0 \\ 0 \\ 0 \\ \frac{1}{A_T}\omega_1 \end{pmatrix} \tag{23}$$

3. Window opening

When the window is opening, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{4\omega_0}{A_c \cos\frac{\pi}{4} + \frac{\theta}{2} l} \\ 0 \\ 0 \end{pmatrix} \tag{24}$$

4. Window closing

When the window is closing, the Δ terms that are added are:

$$\Delta = \begin{pmatrix} \dot{T}_r \\ \dot{T}_2 \\ \dot{T}_B \\ \dot{\theta} \\ \ddot{\theta} \\ \dot{h} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \ddot{\theta} dt \\ \left(-\frac{l}{2} mg \sin\theta \right. \\ + \left(A_c^2 R_0 \frac{1}{4} l \cos(\frac{\pi}{4} + \frac{\theta}{2})\dot{\theta} \right. \\ \left. \left. + A_c P_0\right) \sin(\frac{\pi}{4} - \frac{\theta}{2})\right)/I \\ 0 \end{pmatrix} \tag{25}$$

5. Simulation

The model is simulated in python, with all parameters as defined in appendix B

6. Results

The simulation was conducted under various conditions, and the corresponding plots are available in C. Figure 5 displays a basic heating mode simulation in which the boiler heats water, subsequently warming the radiator and room. As can be observed in the figure, the boiler cools down during periods of sunlight absence, resulting in a decrease in temperature.

The cooling system can be simulated as well, by switching valves 1 and 2. The room cools down, which can be observed in figure C.

II. Sensing and Control model

1. Markov chain

The behaviour of the electric heater can be modelled with the a Markov chain. This chain is shown in figure 1.

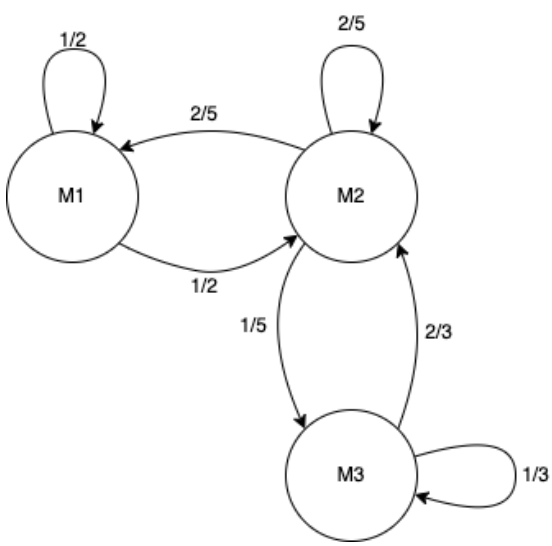

Figure 1. Markov chain of electric heating machine

2. Temperature Sensor

The logic of the temperature sensor can be captured on with the Moore machine shown in figure 2. The output of the sensor is a single integer which can be used as input for the controller of the climate control system.

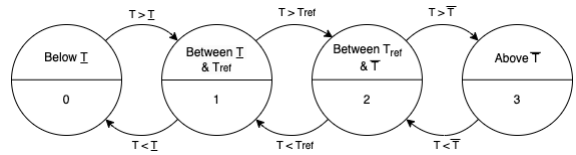

Figure 2. Moore machine of temperature sensor

3. Controller Logic Design

A controller that changes the heating/cooling modes based on the temperature is to be designed. The inputs of the control system are the measurements the sensors take on the temperature of the room and the states of the valves. The controller should change the heating mode to four different states, based on four ranges of the temperature of the room. The four main heating system modes are displayed in Table 1.

domain of T_r	signal of T_r	heating/cooling mode
$T_r < \underline{T}$	0	window is closed, heating is on
$\underline{T} \leq T_r < T_{ref}$	1	window is closed, heating/cooling is off
$T_{ref} \leq T_r < \bar{T}$	2	window is open, heating/cooling is off
$\bar{T} \leq T_r$	3	window is closed, cooling is on

Table 1. Heating system modes

As the controller should produce the output that controls the valves and the pumps based on the input of the measurements, the output and input signals should be defined. First, the measurement signals are defined with respect to the taken measurement. The signal of the temperature measurements are not restated, but it should be noted that they are part of the input as well.

Based on Table 1 and Table 2 the input alphabet of the controller is established to be $\Sigma = \{-1, 0, 1, 2, 3\}$.

As the next step the output of the controller is defined.

subject	state	signal
v_0	window opening	1
v_0	window not moving	0
v_0	window closing	-1
v_1, v_2	heating mode	1
v_1, v_2	disabled heating/cooling	0
v_1, v_2	cooling mode	-1

Table 2. Measurement signals

Five entities are controlled in the simulation: the three valves and the two pumps present. Since it is indicated that it takes time to change the state of the valves we have chosen to control the engines that rotate the valves to set up a coherent simulation. The behaviour of the engines that control the valve with respect to their input signal is displayed on Table 3. The same information for the pumps is presented in Table 4

control signal	change inflicted in the valve
-1	change towards -1 state
0	no change
1	change towards 1 state

Table 3. Valve control

control signal	state of the pump
0	turned off
1	turned on

Table 4. Pump control

Thus the output alphabet of the controller is defined to be $\Gamma = \{-1, 0, 1\}$

The input and output signal of the controller is defined as a tuple, where the input signal is: (T_r, v_0, v_1, v_2, w) so the room temperature, states of valve 0, 1 and 2, and the state of the window. The output signal is: $(c_0, c_1, c_2, c_{p0}, c_{p1})$ so the controller of the engine turning valve 0, 1 and 2, and the controller for pump 0 and 1. The set up is also illustrated on Figure 3.

The states of the machine from subsection II.A, about the electric heater, is not incorporated in the

Figure 3. Controller Input and Output Tuples

Moore machine. This is since the behaviour of the electric heater is random, and does not have an influence on which state the control system is working in. It does have an in influence on the room temperature, so it will eventually cause the control system to act upon these changes.

The Moore machine can be found found in appendix D. Because some states are used double, there is a reduction in nodes possible. This version is a bit less intuitively to understand, however, this version is used to create the code for the control system, and is included in appendix E.

III. Hybrid system

1. Hybrid System

With all the answers on the previous exercises, a hybrid system is derived for the six main states; the different temperature states of the controller, and the window opening and closing states. Since the state space equations can be quite bulky, they are refered to as their equation number. So $\dot{x}$ = (20) + (21) shows that $\dot{x}$ is equal to equation 20 and 21.

The Hybrid system can be found in appendix F.

2. Simulation

When the electric heater is outputting 500 watts, the room should maintain a temperature of 27 degrees Celsius. The thermal resistance of the window can be calculated for this:

$$0 = (T_0 - T_r)\frac{1}{R_{open}} - 500 \tag{26}$$

$$R_{open} = \frac{T_0 - T_r}{500}, \ (T_0 = 25, \ T_r = 27) \tag{27}$$

$$R_{open} = \frac{1}{250} \tag{28}$$

This is the value picked for the thermal resistance. Since the heaters does have an effect on the temperature, the following term is added to each state space model in all different states:

$$\Delta T_r = \frac{W_r}{C_r} \tag{29}$$

where W_r is the power of the machine, and C_r is the heat capacity of the room.

The whole system is simulated for 24 hours. As can be observed, the controller keeps the room around 27 degrees Celsius, even tho the electric heater is switching states. The window is closing and opening as well, which influences the temperature. All figures can be found in appendix G.

The system is simulated with a constant T_0, but also more realistically using a varying T_0 where it is represented by a sin function with an amplitude of 10 C° and mean of 25 C°. For both instances a simulation with and without the electric heater is run.

IV. Appendix

A. Bond Graph

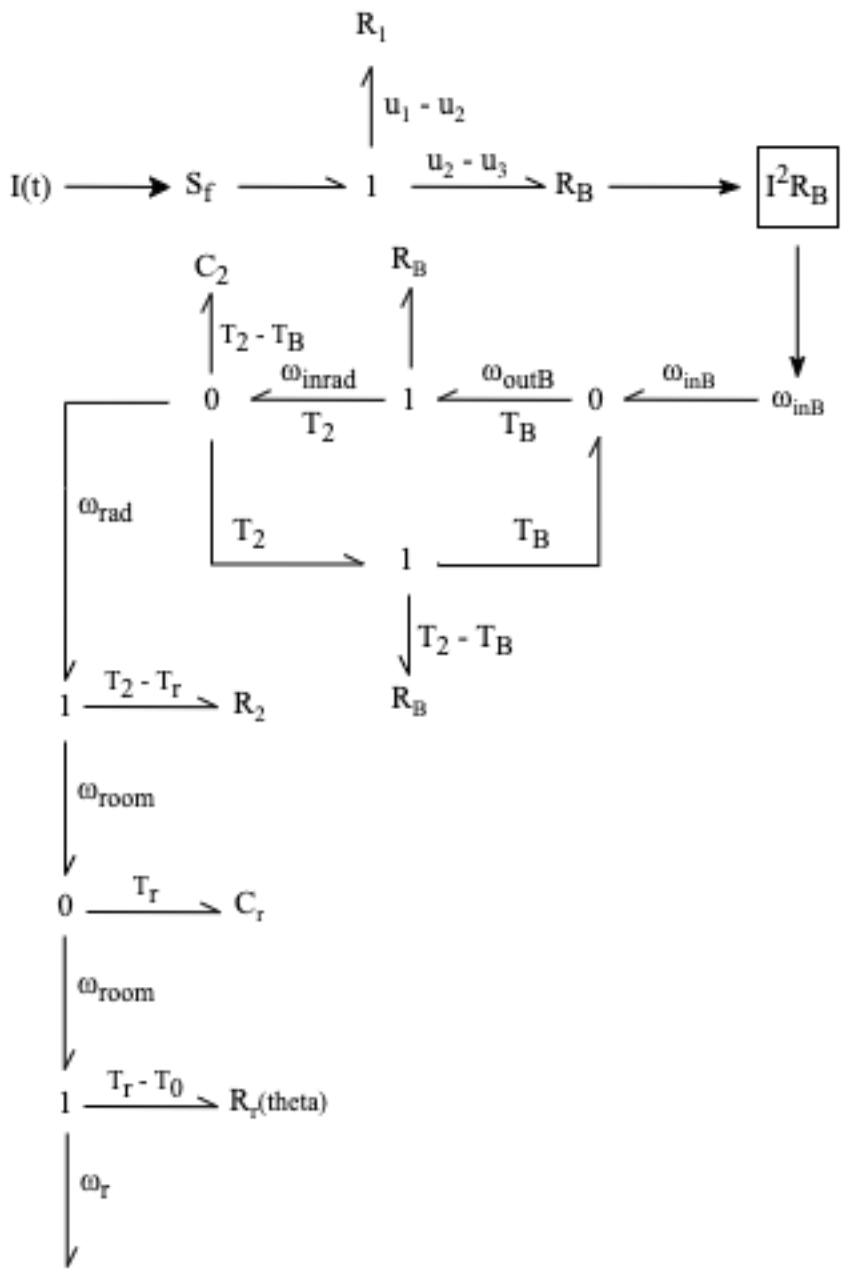

Figure 4. Bond Graph

B. Parameters

Parameter	Value
C_r	800,000
C_2	200,000
C_b	200,000
R_2	0.02
R_B	5
R_1	0.01
T_1	275
T_0	299
$\hat{R}_0$	0.004
$\hat{R}$	0.1
ρ	1,000 kg/m^3
g	9.81 m/s^2
l	1.4 m
$\omega 0$	0.000002 l/s
$\omega 1$	0.1 l/s
A_t	0.01 m^2
A_c	0.01 m^2
m	1 kg
R_0	5,000,000
R_t	5,000,000
dt_big	1
dt_small	1
Amp	400
ω_rad	$2\pi/(2 \times 24 \times 60 \times 60)$
A_s	5 m^2
μ	0.1
P_0	10^5 Pa
$t1$	21,600 s
$t2$	72,000 s
$f1$	14.8

Table 5. List of Parameters

C. Simulation results

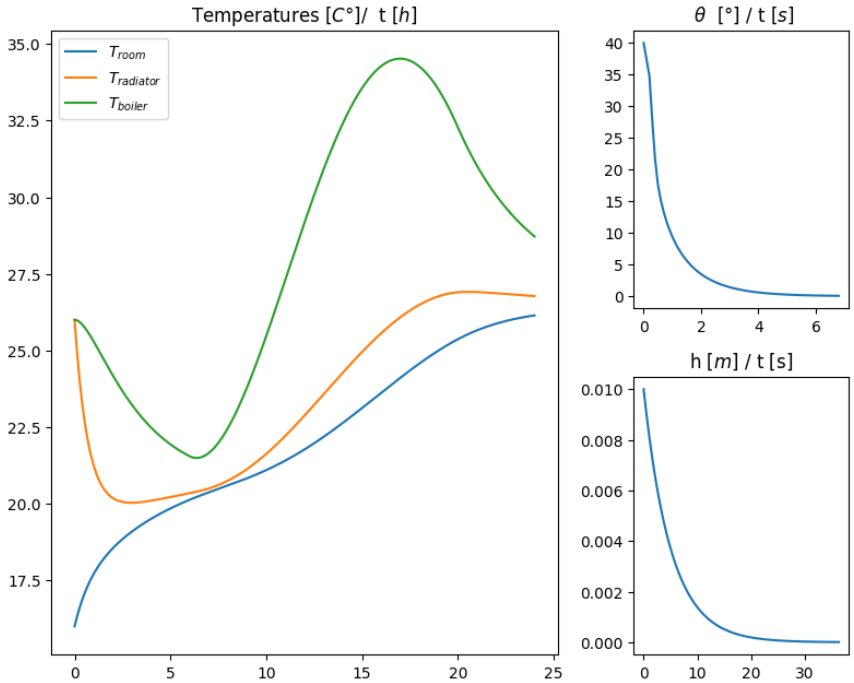

Figure 5. Simulation of the heating system, power by the solar panel.

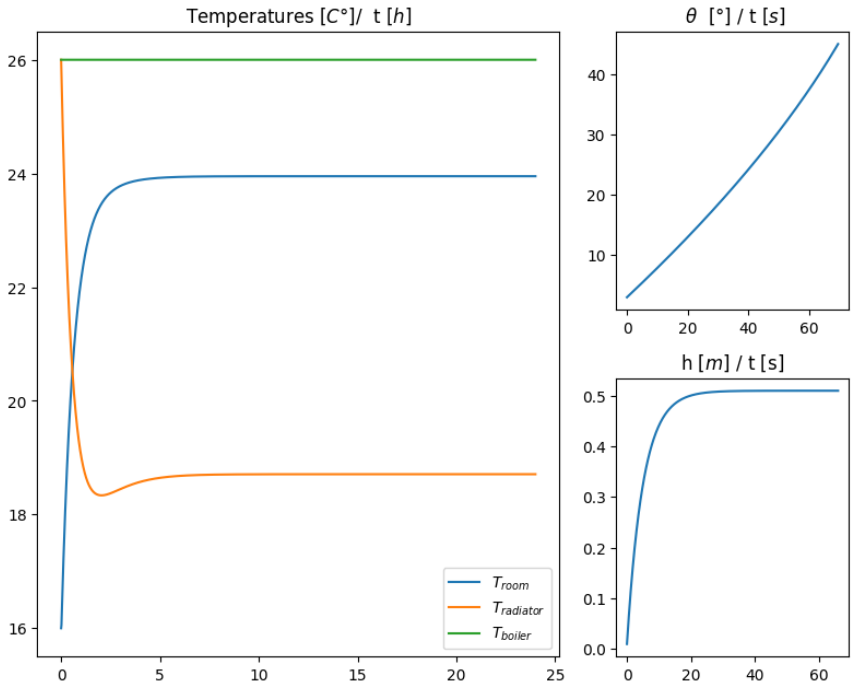

Figure 6. Simulation of the cooling system, with a the window opening.

D. Moore machine controller - extended

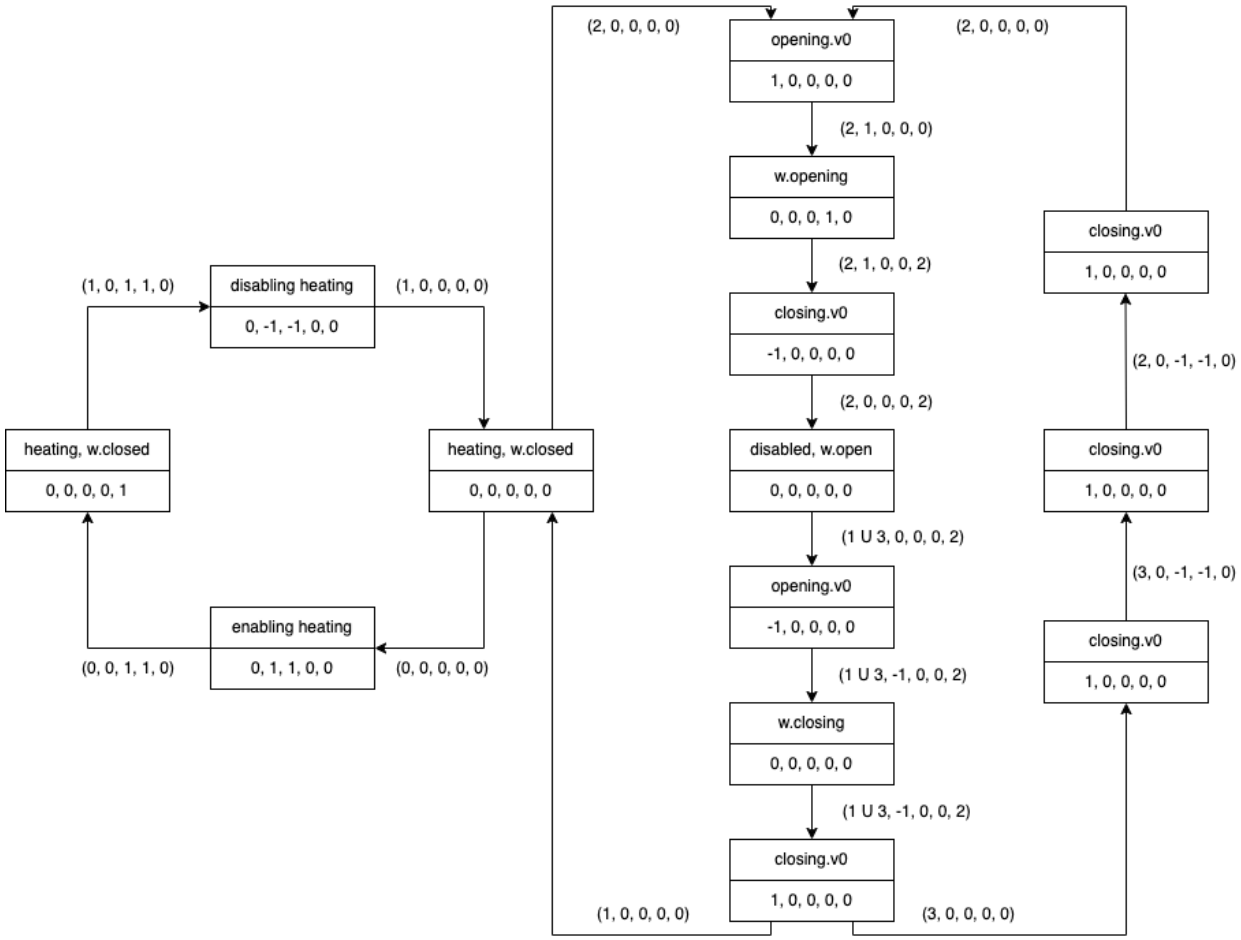

Figure 7. Moore machine of extended controller

E. Moore machine controller - reduced

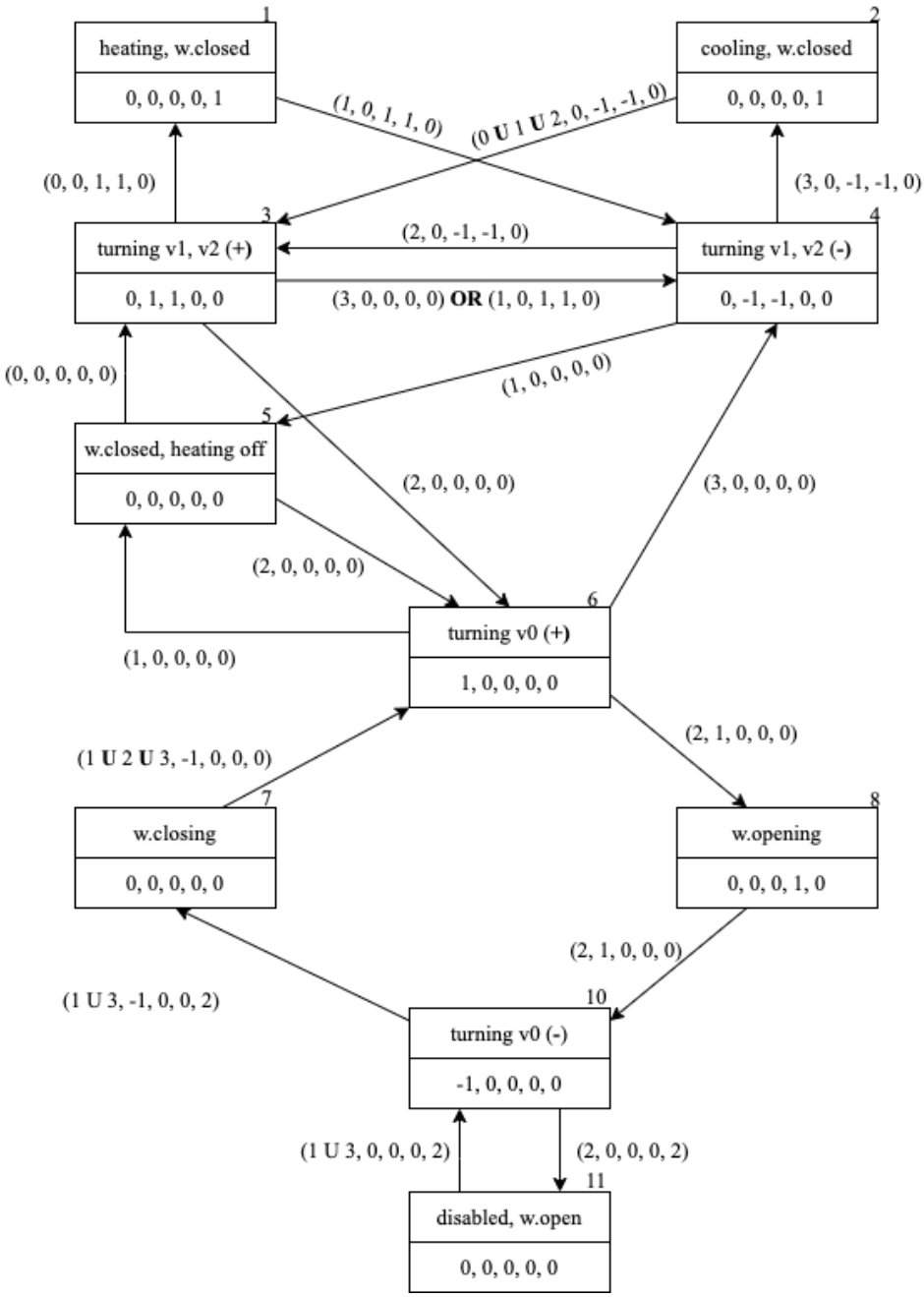

Figure 8. Moore machine of controller with reduced amount of nodes

F. Hybrid System

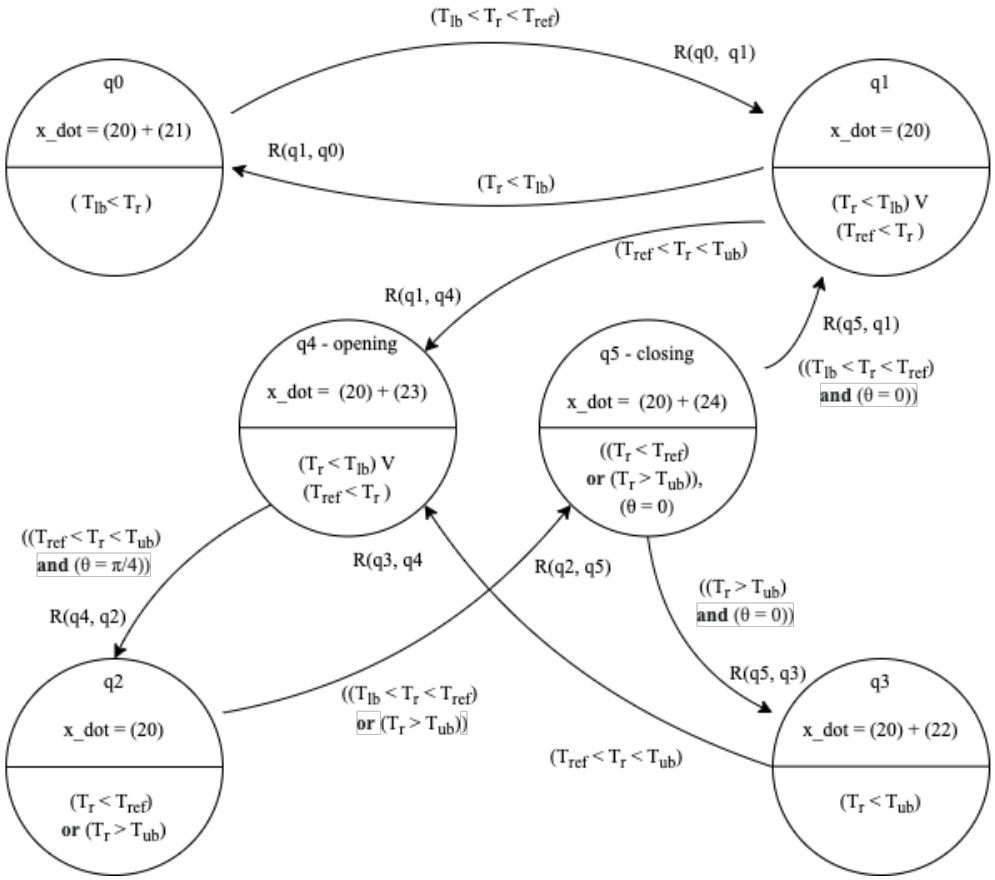

Figure 9. The Hybrid system diagram

G. Simulation Results

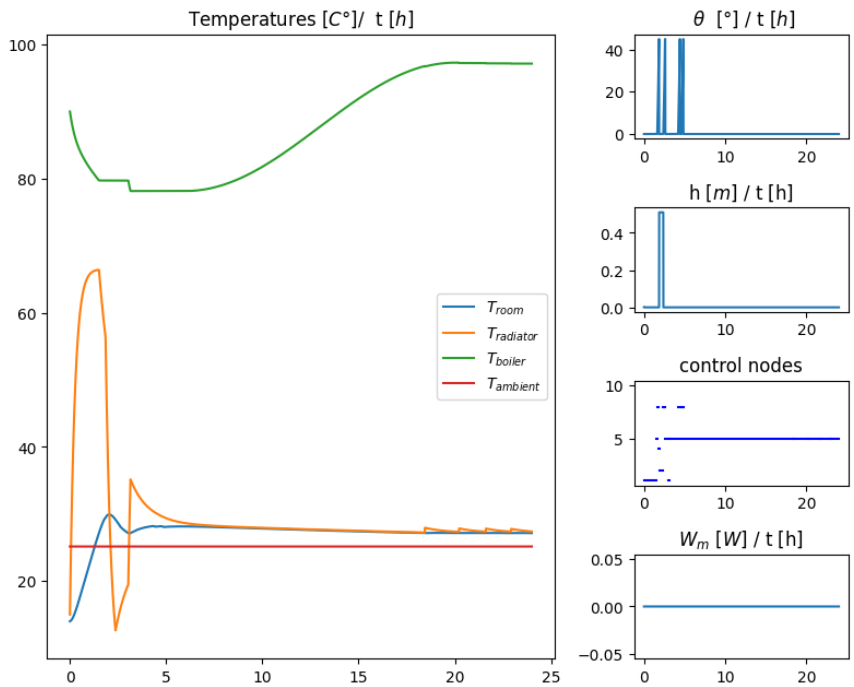

Figure 10. Simulation with a constant T_0, without the heater switching modes.

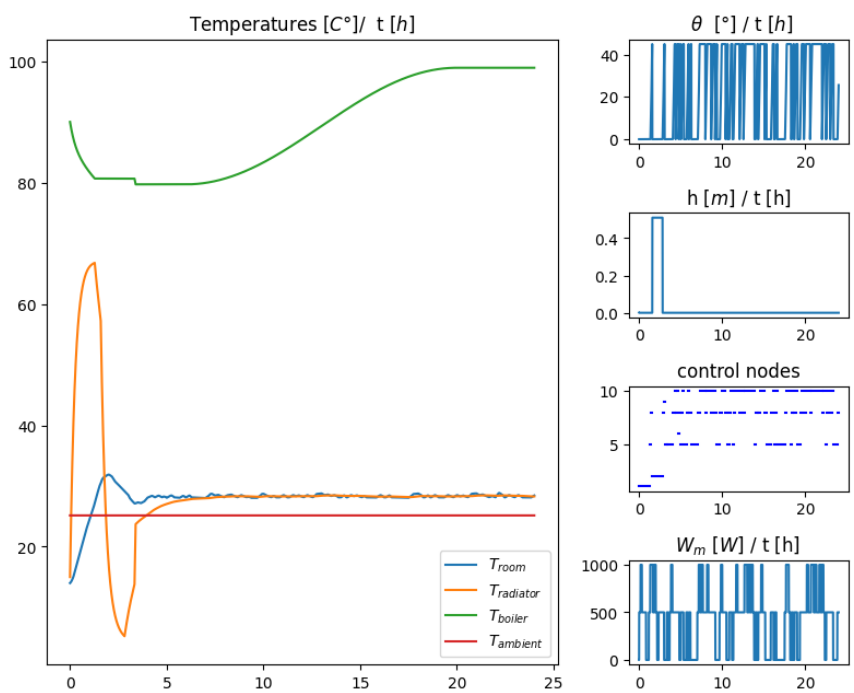

Figure 11. Simulation with a constant T_0, with the heater switching modes.

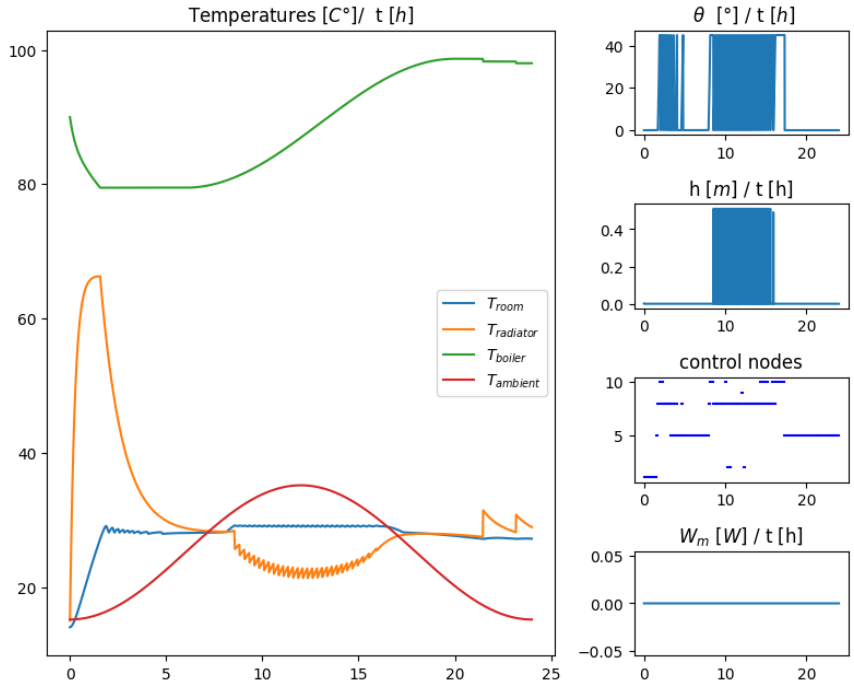

Figure 12. Simulation with a varying T_0, without the heater switching modes.

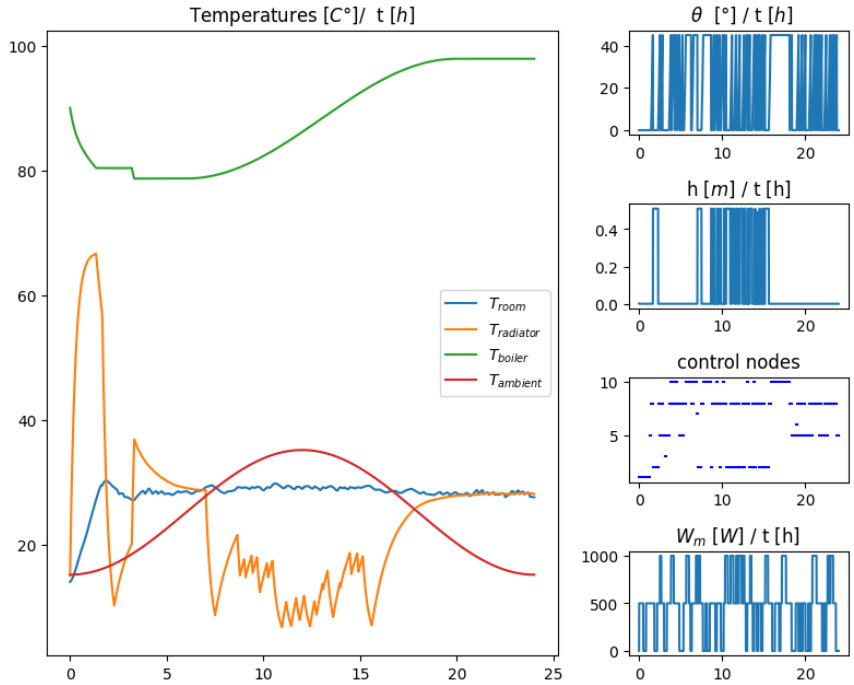

Figure 13. Simulation with a varying T_0, with the heater switching modes.